

Lab 4: MATLAB Programming

Authors	Name: 何衍宁 Name: 夏天	Student ID:12311512 Student ID:12311610
Objective Lab4 In Lab 4, our primary learning objectives are centered around deepening our understanding of frequency-domain analysis through the use of the continuous-time Fourier transform (CTFT). We aim to develop a strong grasp of how to analyze both periodic and aperiodic signals in the frequency domain, which simplifies the structure of complex time-domain signals. By engaging with various exercises, we will enhance our intuitive understanding of the relationship between time-domain and frequency-domain properties, particularly in the context of linear time-invariant (LTI) systems. Additionally, we will explore practical applications such as using frequency response analyses to understand system behaviors and applying numerical methods to approximate the CTFT. This lab will not only strengthen our theoretical knowledge but also improve our practical skills in handling real-world signal processing challenges. Lab results & Analysis 4.2 ■ 4.2 Numerical Approximation to the Continuous-Time Fourier Transform A large class of signals can be represented using the continuous-time Fourier transform (CTFT) in Eq. (4.1). In this exercise you will use MATLAB to compute numerical approximations to the CTFT integral, Eq. (4.2). By approximating the integral using a summation over closely spaced samples in t, you will be able to use the function <code>fft</code> to compute your approximation very efficiently. The approximation you will use follows from the definition of the integral $\int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = \lim_{\tau \rightarrow 0} \sum_{n=-\infty}^{\infty} x(n\tau)e^{-j\omega n\tau}.$ For a large class of signals and for sufficiently small τ, the sum on the right-hand side is a good approximation to the CTFT integral. If the signal $x(t)$ is equal to zero for $t < 0$ and		

$t \geq T$, then the approximation can be written

$$\int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = \int_0^T x(t)e^{-j\omega t} dt \approx \sum_{n=0}^{N-1} x(n\tau)e^{-j\omega n\tau}, \quad (4.7)$$

where $T = N\tau$ and N is an integer. You can use the function `fft` to compute the sum in Eq. (4.7) for a discrete set of frequencies ω_k . If the N samples $x(n\tau)$ are stored in the vector `x` then the function call `X=tau*fft(x)` calculates

$$X(k+1) = \tau \sum_{n=0}^{N-1} x(n\tau)e^{-j\omega_k n\tau}, \quad 0 \leq k < N, \quad (4.8)$$

$$\approx X(j\omega_k), \quad (4.9)$$

where

$$\omega_k = \begin{cases} \frac{2\pi k}{N\tau}, & 0 \leq k \leq \frac{N}{2}, \\ \frac{2\pi k}{N\tau} - \frac{2\pi}{\tau}, & \frac{N}{2} + 1 \leq k < N, \end{cases} \quad (4.10)$$

and N is assumed to be even. For reasons of computational efficiency, `fft` returns the positive frequency samples before the negative frequency samples. To place the frequency samples in ascending order, you can use the function `fftshift`. To store in `X` samples of $X(j\omega_k)$ ordered such that `X(k+1)` is the CTFT evaluated at $-\pi/\tau + (2\pi k/N\tau)$ for $0 \leq k \leq N-1$, use `X=fftshift(tau*fft(x))`.

For this exercise, you will approximate the CTFT of $x(t) = e^{-2|t|}$ using the function `fft` and a truncated version of $x(t)$. You will see that for sufficiently small τ , you can compute an accurate numerical approximation to $X(j\omega)$.

4.2 a

Basic Problems

(a). Find an analytic expression for the CTFT of $x(t) = e^{-2|t|}$. You may find it helpful to think of $x(t) = g(t) + g(-t)$, where $g(t) = e^{-2t}u(t)$.

Find the analytic CTFT expression for $x(t) = \exp(-2 \cdot \text{abs}(t))$

Analysis:

Think of $x(t)$ as $\exp(-2 \cdot t) \cdot u(t) + \exp(2 \cdot t) \cdot u(-t)$.

This question needs no code.

Lab result:

$$X(j\omega) = 1/(2+j\omega) + 1/(2-j\omega)$$

4.2 b

(b). Create a vector containing samples of the signal $y(t) = x(t - 5)$ for $\tau = 0.01$ and $T = 10$ over the range $t = [0:\tau:T-\tau]$. Since $x(t)$ is effectively zero for $|t| > 5$, you can calculate the CTFT of the signal $y(t) = x(t - 5)$ from the above analysis using $N = T/\tau$. Your vector `y` should have length N .

Let $y(t) = x(t-5)$ and calculate the CTFT expression for $y(t)$, where $\tau=0.01$ and $T=10$

Analysis:

$N = T/\tau$, when N is even, use `fft` method; At the same time, find the value of $|Y(j\omega)|$ and plot

```
clear;

clc;

t = 0:0.01:10-0.01

tau = 0.01;

N = 10/tau;

y = exp(-2*abs((t-5)))

Y_temp = tau*fft(y)

w = -pi/tau + (0:N-1)*2*pi/(N*tau)

Xa = 1./(2+1j.*w) + 1./(2-1j.*w);

Ya = exp(-1j*5*w).*Xa;

subplot(2,1,1),

plot(t,y);

xlabel('t'),

ylabel('y(t)'),

legend('y(t)');

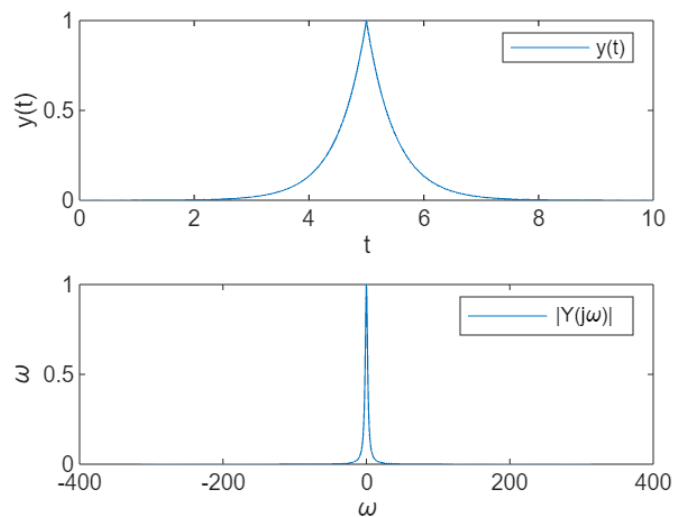
subplot(2,1,2),

plot(w,abs(Ya));

xlabel('\omega'),

ylabel('\omega'),

legend('|Y(j\omega)|');
```



Lab result:

The $Y(j\omega)$ is like the figure.

4.2 c

(c). Calculate samples $Y(j\omega_k)$ by typing $Y = \text{fftshift}(\text{tau} * \text{fft}(y))$.

Use `fftshift` method to transform Y

Analysis:

Use the `fftshift` method

```
t = 0:0.01:10-0.01
```

```
tau = 0.01;
```

```
N = 10/tau;
```

```
y = exp(-2*abs((t-5)))
```

```
Y = fftshift(tau*fft(y))
```

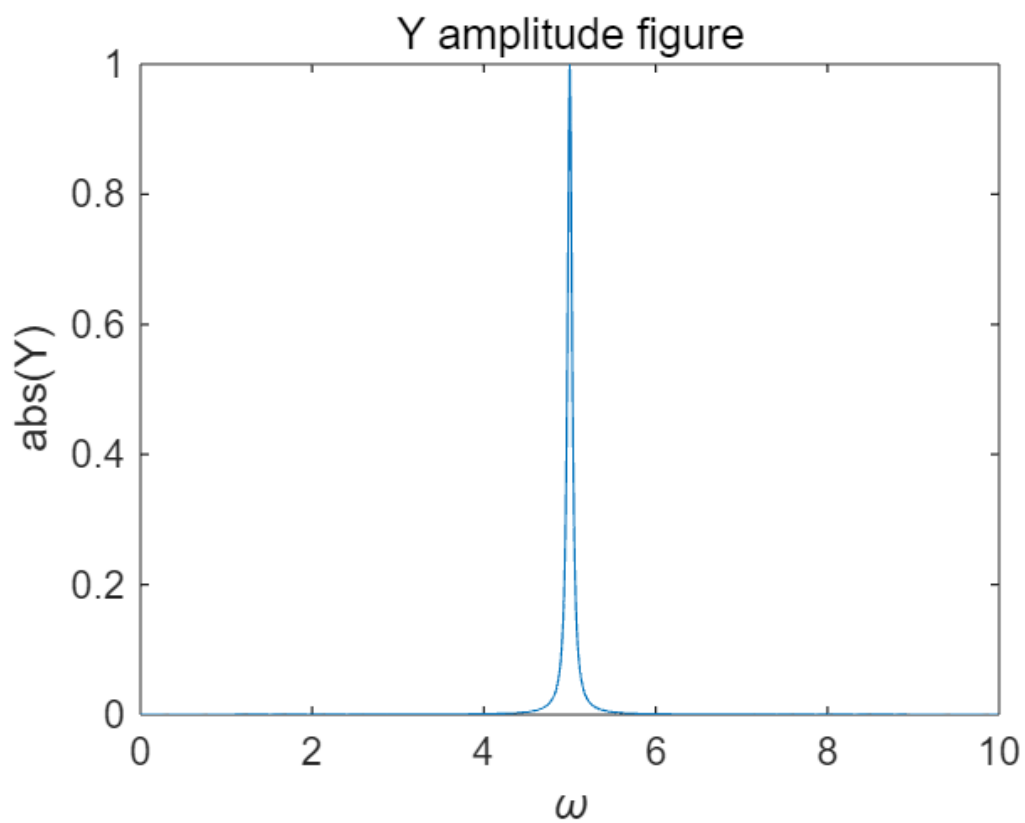
```
figure
```

```
plot(t,abs(Y));
```

```
title('Y amplitude figure '),
```

```
xlabel('\omega'),
```

```
ylabel('abs(Y)');
```



Lab result:

$Y(j\omega)$ is like the figure.

4.2 d

(d). Construct a vector w of frequency samples that correspond to the values stored in the vector Y as follows

```
>> w = -(pi/tau)+(0:N-1)*(2*pi/(N*tau));
```

Determine the range w of the frequency sample

Analysis:

We have the formula

```
clear;

clc;

t = 0:0.01:10-0.01;

tau = 0.01;

N = 10/tau;

y = exp(-2*abs((t-5)));

w = -pi/tau + (0:N-1)*2*pi/(N*tau)
```

```
w = 1×1000
-314.1593 -313.5309 -312.9026 -312.2743 -311.6460 -311.0177 -310.3894
-309.7610 -309.1327 -308.5044 -307.8761 -307.2478 -306.6194 -305.9911
-305.3628 -304.7345 -304.1062 -303.4779 -302.8495 -302.2212 -301.5929
-300.9646 -300.3363 -299.7079 -299.0796 -298.4513 -297.8230 -297.1947
-296.5663 -295.9380 -295.3097 -294.6814 -294.0531 -293.4248 -292.7964
-292.1681 -291.5398 -290.9115 -290.2832 -289.6548 -289.0265 -288.3982
-287.7699 -287.1416 -286.5133 -285.8849 -285.2566 -284.6283 -284.0000
-283.3717
```

Lab result:

We find the ω .

4.2 e

(e). Since $y(t)$ is related to $x(t)$ through a time shift, the CTFT $X(j\omega)$ is related to $Y(j\omega)$ by a linear phase term of the form $e^{j5\omega}$. Using the frequency vector ω , compute samples of $X(j\omega)$ directly from Y , storing the result in the vector X .

The time shift corresponds to the frequency domain shift to calculate X from Y

Analysis:

Time shift 5, then the frequency domain is multiplied by $\exp(j*5*\omega)$.

```
clear;

clc;

t = 0:0.01:10-0.01;

tau = 0.01;

N = 10/tau;

y = exp(-2*abs((t-5)));

w = -pi/tau + (0:N-1)*2*pi/(N*tau);

Y = fftshift(tau*fft(y));

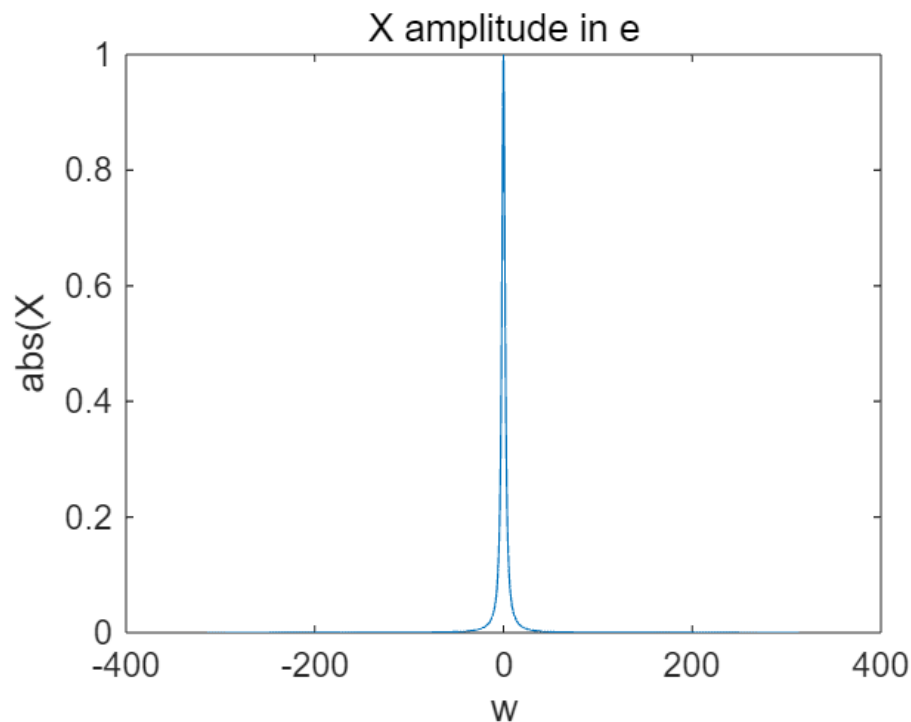
X = Y.*exp(1j.*5*w);

plot(w,abs(X));

title('X amplitude in e'),

xlabel('w'),

ylabel('abs(X');
```



Lab result:

Figure is drawn.

4.2 f

- (f). Using `abs` and `angle`, plot the magnitude and phase of X over the frequency range specified in `w`. For the same values of ω , also plot the magnitude and phase of the analytic expression you derived for $X(j\omega)$ in Part (a). Does your approximation of the CTFT match what you derived analytically? If you plot the magnitude on a logarithmic scale, using `semilogy`, you will notice that at higher frequencies the approximation is not as good as at lower frequencies. Since you have approximated $x(t)$ with samples $x(n\tau)$, your approximation will be better for frequency components of the signal that do not vary much over time intervals of length τ .

Draw the amplitude and Angle plots of $X(j\omega)$ in question a and $X(j\omega)$ in question e respectively

Analysis:

Use `w` as the range of horizontal coordinates; The difference of X amplitude in a/e problem is compared by using `semilogy` function of `semilog` graph.

```
clear;

clc;

t = 0:0.01:10-0.01;
```

```
tau = 0.01;

N = 10/tau;

y = exp(-2*abs((t-5)));

w = -pi/tau + (0:N-1)*2*pi/(N*tau);

Y = fftshift(tau*fft(y));

X = Y.*exp(1j.*5*w);

figure

subplot(2,2,1),

plot(w,abs(X));

title('X magnitude in e'),

xlabel('w'),

ylabel('abs(X)');

subplot(2,2,2),

plot(w,angle(X));

title('X phase in e'),

xlabel('w'),

ylabel('phase(X)');

X1 = 1./(2+1j.*w) + 1./(2-1j.*w);

subplot(2,2,3),

plot(w,abs(X1));

title('X magnitude in a'),

xlabel('w'),

ylabel('abs(X)');

subplot(2,2,4),
```

```
plot(w,angle(X1));

title('X phase in a'),

xlabel('w'),

ylabel('phase(X)');


figure

subplot(1,2,1)

plot(w,abs(X))

title('X magnitude in e')

ylabel('abs(X)-e')

xlabel('ω')

ylim([0.001,1])

subplot(1,2,2)

plot(w,abs(X1))

title('X magnitude in a')

ylabel('abs(X)-a')

xlabel('ω')

xlim([-20,20])

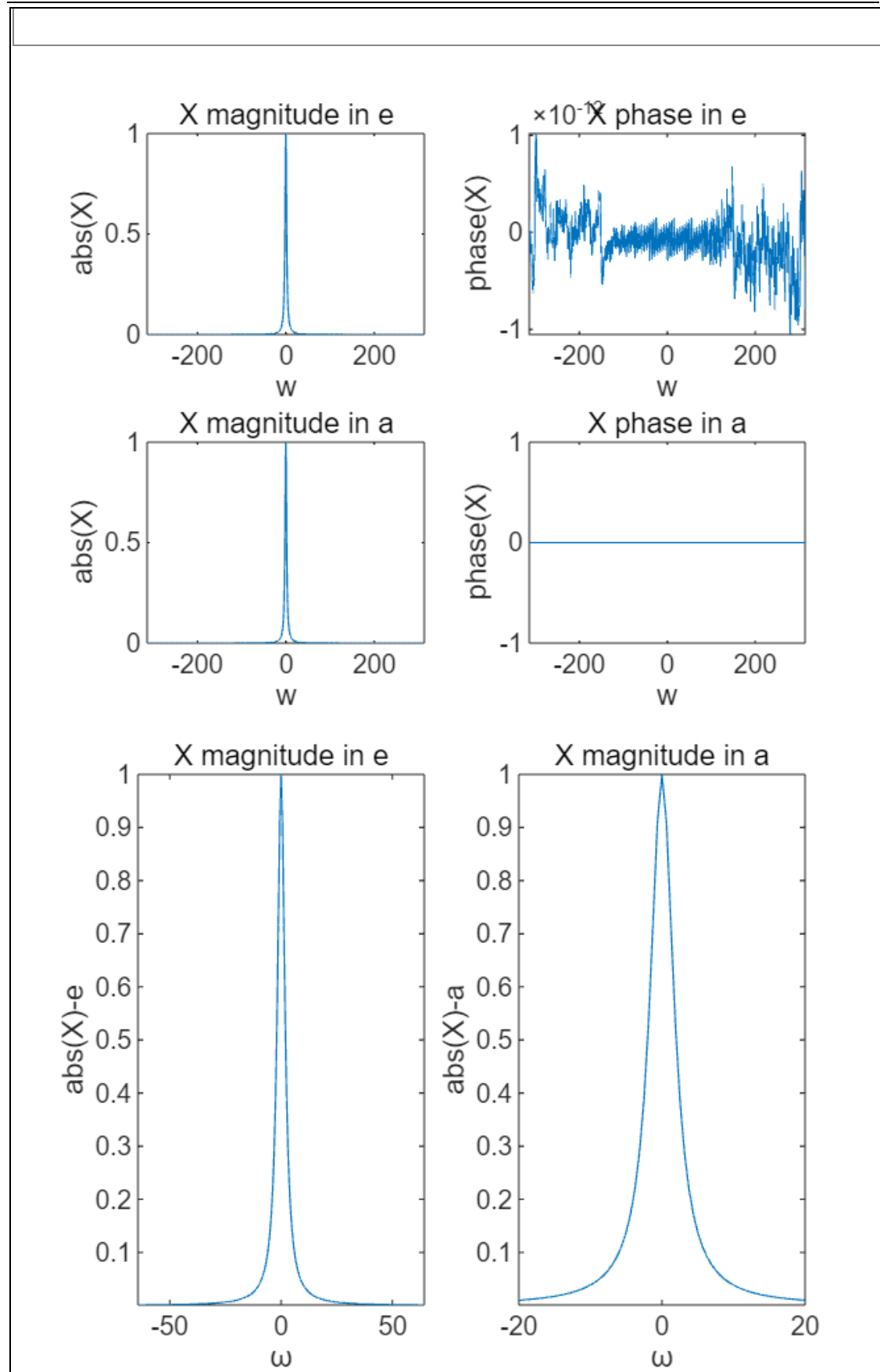
ylim([0.001,1])

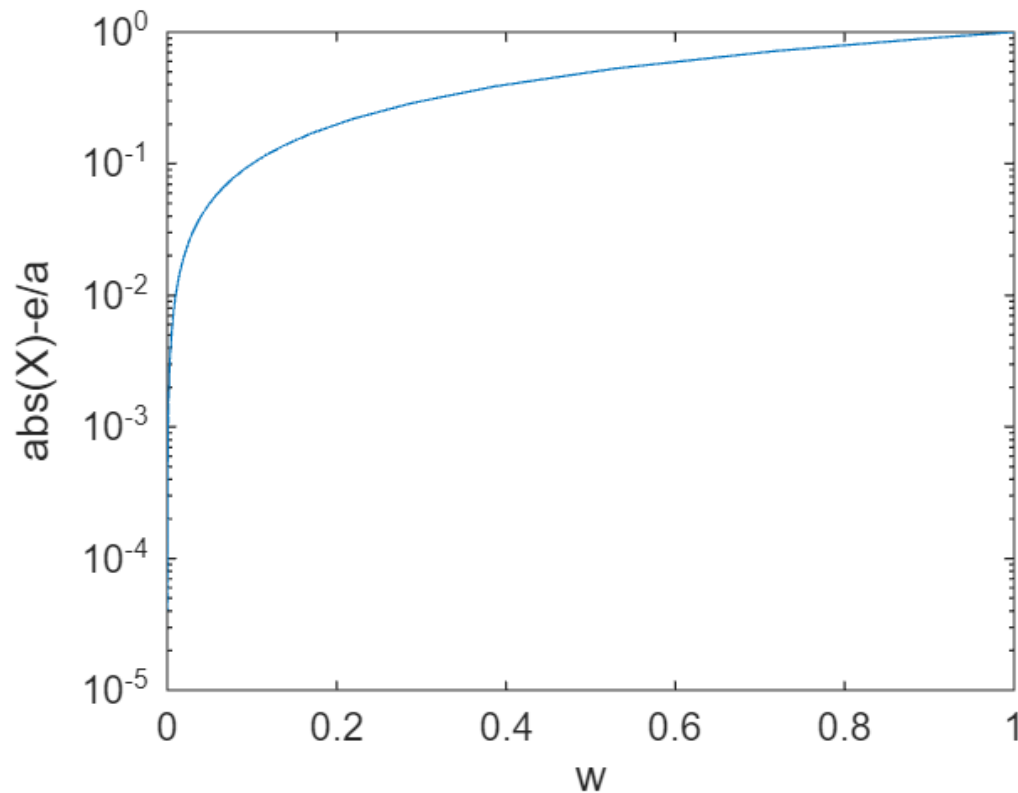

figure

semilogy(abs(X),abs(X1))

xlabel('w'),

ylabel('abs(X)-e/a')
```





Lab result:

It can be seen from the figure **abs(X)-e** and **abs(X)-a** that the approximate value of CTFT is consistent with the answer obtained by analysis, but there are still differences in the specific value. And the amplitude and phase of X are shown in the first four figures. The last figure is about magnitude on a logarithmic scale.

4.2 g

(g). Plot the magnitude and phase of Y using abs and angle. How do they compare with X? Could you have anticipated this result?

Draw the amplitude and phase Angle plots for Y and compare them with X

Analysis:

The time domain signal is time-shifted, and the corresponding frequency domain signal amplitude is unchanged, and the phase Angle is changed

```
clear;

clc;

t = 0:0.01:10-0.01;

tau = 0.01;

N = 10/tau;

y = exp(-2*abs((t-5)));

w = -pi/tau + (0:N-1)*2*pi/(N*tau);

Y = fftshift(tau*fft(y));

X = Y.*exp(1j.*5*w);

figure

subplot(2,2,1),

plot(w,abs(X));

title('X magnitude'),

xlabel('w'),

ylabel('abs(X)');

subplot(2,2,2),

plot(w,angle(X));

title('X phase'),

xlabel('w'),

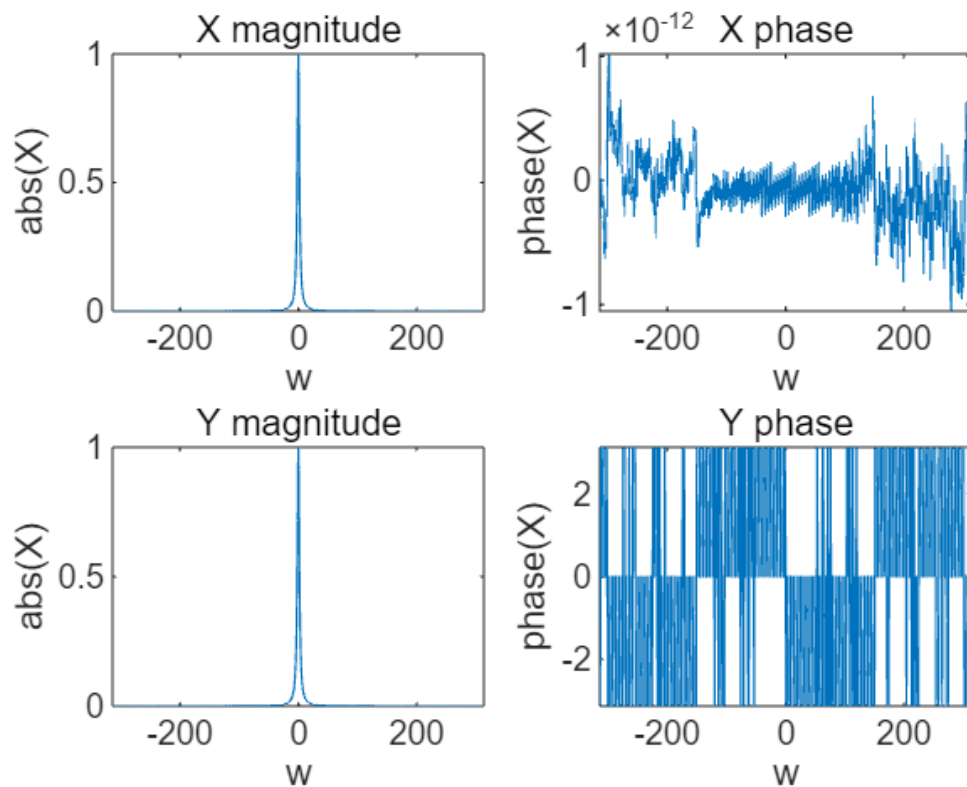
ylabel('phase(X)');

subplot(2,2,3),

plot(w,abs(Y));

title('Y magnitude'),
```

```
xlabel('w'),  
  
ylabel('abs(X)');  
  
subplot(2,2,4),  
  
plot(w,angle(Y));  
  
title('Y phase'),  
  
xlabel('w'),  
  
ylabel('phase(X)');
```



Lab result:

The time domain signal is time-shifted, and the corresponding frequency domain signal amplitude is unchanged, and the phase Angle is changed. As shown in the figure, the amplitude diagram is the same, but the phase Angle diagram is different.

4.6

4.6 Amplitude Modulation and the Continuous-Time Fourier Transform

This exercise will explore amplitude modulation of Morse code messages. A simple amplitude modulation system can be described by

$$x(t) = m(t) \cos(2\pi f_0 t), \quad (4.13)$$

where $m(t)$ is called the message waveform and f_0 is the modulation frequency. The continuous-time Fourier transform (CTFT) of a cosine of frequency f_0 is

$$C(j\omega) = \pi\delta(\omega - 2\pi f_0) + \pi\delta(\omega + 2\pi f_0), \quad (4.14)$$

which can be confirmed by substituting $C(j\omega)$ into Eq. (4.1) to yield

$$\cos(2\pi f_0 t) = \frac{1}{2} (e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}). \quad (4.15)$$

Using $C(j\omega)$ and the multiplication property of the CTFT, you can obtain the CTFT of $x(t)$,

$$X(j\omega) = \frac{1}{2} M(j(\omega - 2\pi f_0)) + \frac{1}{2} M(j(\omega + 2\pi f_0)), \quad (4.16)$$

where $M(j\omega)$ is the CTFT of $m(t)$. Since the CTFT of a sinusoid can be expressed in terms of impulses in the frequency domain, multiplying the signal $m(t)$ by a cosine places copies of $M(j\omega)$ at the modulation frequency.

The remainder of this exercise will involve the signal,

$$x(t) = m_1(t) \cos(2\pi f_1 t) + m_2(t) \cos(2\pi f_2 t) + m_3(t) \sin(2\pi f_1 t), \quad (4.17)$$

and several parameters that can be loaded into MATLAB from the file `ctftmod.mat`. This file is in the Computer Explorations Toolbox, which can be obtained from The MathWorks at the address listed in the Preface. If the file is in one of the directories in your MATLABPATH, type `load ctftmod.mat` to load the required data. The directories contained in your MATLABPATH can be listed by typing `path`. If the file has been successfully loaded, then typing `who` should produce the following result:

```
>> who
```

Your variables are:

```
af      dash      f1      t
bf      dot       f2      x
```

In addition to the signal $x(t)$, you also have loaded:

- a lowpass filter, whose frequency response can be plotted by `freqs(bf,af)`,
- modulation frequencies `f1` and `f2`,
- two prototype signals `dot` and `dash`,
- a sequence of time samples `t`.

To make this exercise interesting, the signal $x(t)$ contains a simple message. When loading the file, you should have noticed that you have been transformed into Agent 008, the code-breaking sleuth. The last words of the aging Agent 007 were "The future of technology lies in ..." at which point Agent 007 produced a floppy disk and keeled over. The floppy disk contained the MATLAB file `ctftmod.mat`. Your job is to decipher the message encoded in $x(t)$ and complete Agent 007's prediction.

Here is what is known. The signal $x(t)$ is of the form of Eq. (4.17), where f_1 and f_2 are given by the variables `f1` and `f2`, respectively. It is also known that each of the signals $m_1(t)$, $m_2(t)$ and $m_3(t)$ correspond to a single letter of the alphabet which has been encoded using International Morse Code, as shown in the following table:

A	.-	H		O	---	V	...-
B	-...	I	..	P		W	..--
C	-.-.	J	---.	Q		X	..--
D	-..	K	-.--	R	..-	Y	..--
E	..	L	..--	S	...	Z	--..
F	...-	M	--	T	-		
G	---	N	..	U	..		

4.6 a

Basic Problems

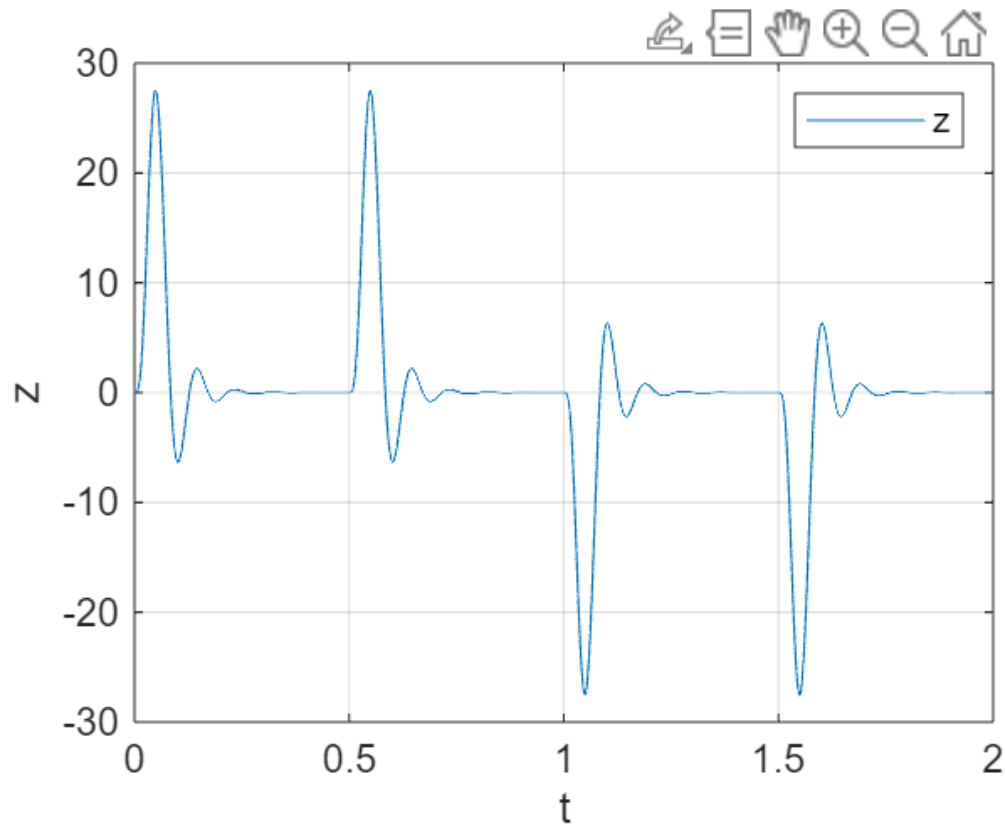
(a). Using the signals dot and dash, construct the signal that corresponds to the letter 'Z' in Morse code, and plot it against t . As an example, the letter C is constructed by typing $c = [\text{dash dot dash dot}]$. Store your signal $z(t)$ in the vector z .

The question requires the use of signals dot and dash to establish a Morse code representing Z and coexist in z , and it is mapped to t :

Analysis:

Import mod files according to the requirements of the topic, and operate according to the requirements of the topic

```
clear;  
  
clc;  
  
load newctftmod.mat  
  
z = [dash,dash,dot,dot];  
  
plot(t,z,'LineStyle','-');  
  
grid on,  
  
xlabel('t'),  
  
ylabel('z'),  
  
legend('z');
```



Lab result:

Figure is drawn.

4.6 b

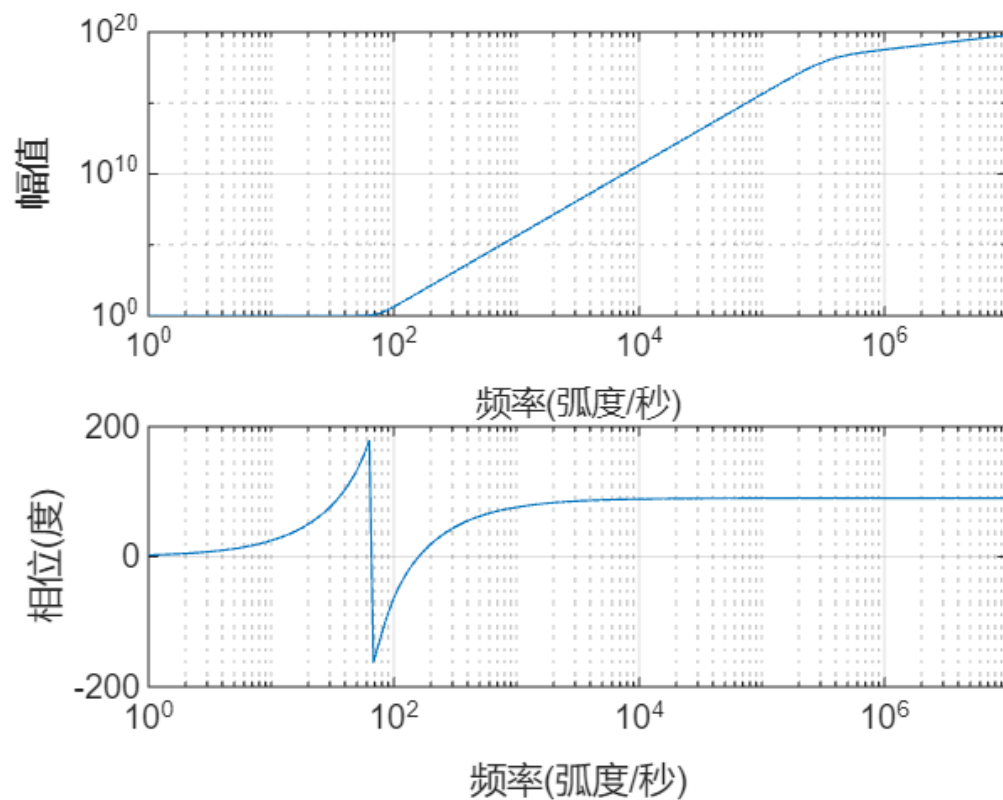
(b). Plot the frequency response of the filter using freqs (bf , af) .

The problem requires the freqs function to make the frequency response of the filter.

Analysis:

Just follow the instructions of the question

```
clear;  
  
clc;  
  
load newctftmod.mat;  
  
freqs(af,bf)
```



Lab result:

Draw the image as shown.

4.6 c

- (c). The signals `dot` and `dash` are each composed of low frequency components such that their Fourier transforms lie roughly within the passband of the lowpass filter. Demonstrate this by filtering each of the two signals using

```
>> ydash=lsim(bf,af,dash,t(1:length(dash)));  
>> ydot=lsim(bf,af,dot,t(1:length(dot)));
```

Plot the outputs `ydash` and `ydot` along with the original signals `dash` and `dot`.

The problem requires filtering `dot` and `dash` signals, making images before and after filtering, and proving that `dot` and `dash` signals can basically pass through Low-pass filter

Analysis:

Filter with `lsim` function filter

```
clear;  
  
clc;  
  
load newctftmod.mat;  
  
ydash = lsim(bf,af,dash,t(1:length(dash)));  
  
ydot = lsim(bf,af,dot,t(1:length(dot)));  
  
figure  
  
subplot(2,1,1)  
  
plot(t(1:length(dash)),dash,t(1:length(dash)),ydash);  
  
xlabel('t'),  
  
ylabel('dash')  
  
grid on;
```

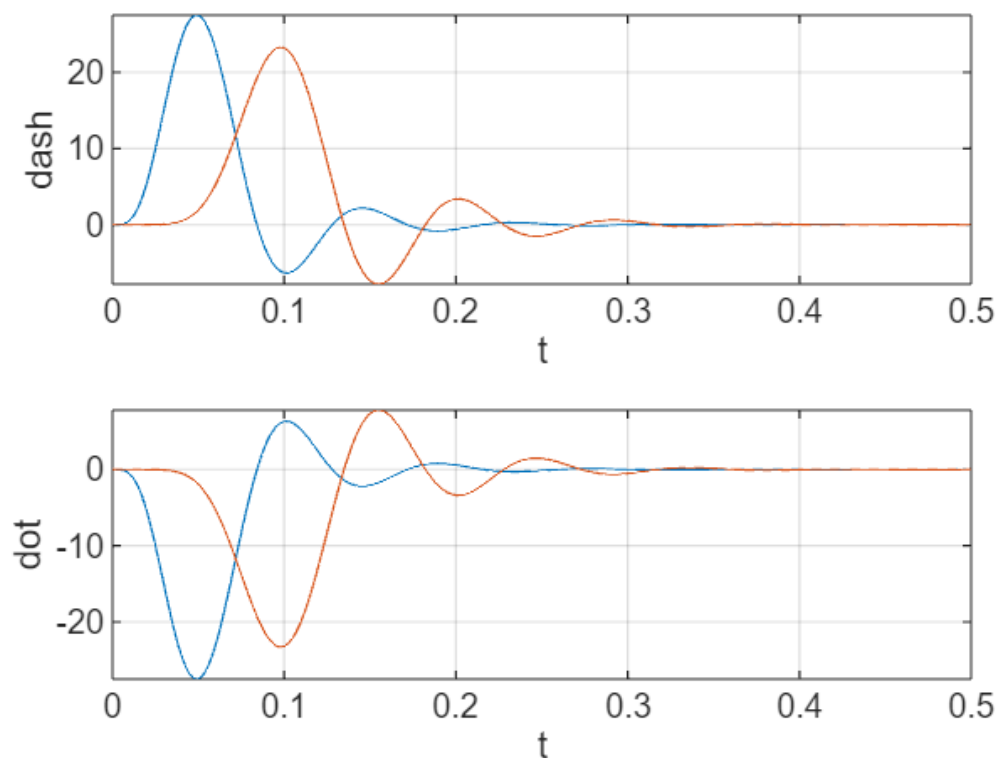
```
subplot(2,1,2)

plot(t(1:length(dot)),dot,t(1:length(dot)),ydot);

xlabel('t'),

ylabel('dot')

grid on;
```

**Lab result:**

Make the image as shown in the figure, you can see that the two signal waveforms are basically unchanged, the energy loss is small, only the phase change, so it can pass the low-pass filter.

4.6 d

(d). When the signal `dash` is modulated by $\cos(2\pi f_1 t)$, most of the energy in the Fourier transform will move outside the passband of the filter. Create the signal $y(t)$ by executing `y=dash*cos(2*pi*f1*t(1:length(dash)))`. Plot the signal $y(t)$. Also plot the output `yo=lsim(bf,af,y,t)`. Do you get a result that you would have expected?

It is required to prove that after the `dash` signal is modulated by $\cos(2\pi f_1 t)$, most of its Fourier transform energy cannot pass through the filter, that is, most of the energy is filtered

Analysis:

Draw y and y_o according to the problem.

```
clear;

clc;

load newctftmod.mat;

ydash = lsim(bf,af,dash,t(1:length(dash)));

ydot = lsim(bf,af,dot,t(1:length(dot)));

y = dash.*cos(2*pi*f1*t(1:length(dash)));

yo = lsim(bf,af,y,t(1:length(y)));

figure

subplot(2,1,1)

plot(t(1:length(dash)),y);

xlabel('t'),

ylabel('y'),

legend('y'),

grid on;
```

```
subplot(2,1,2)

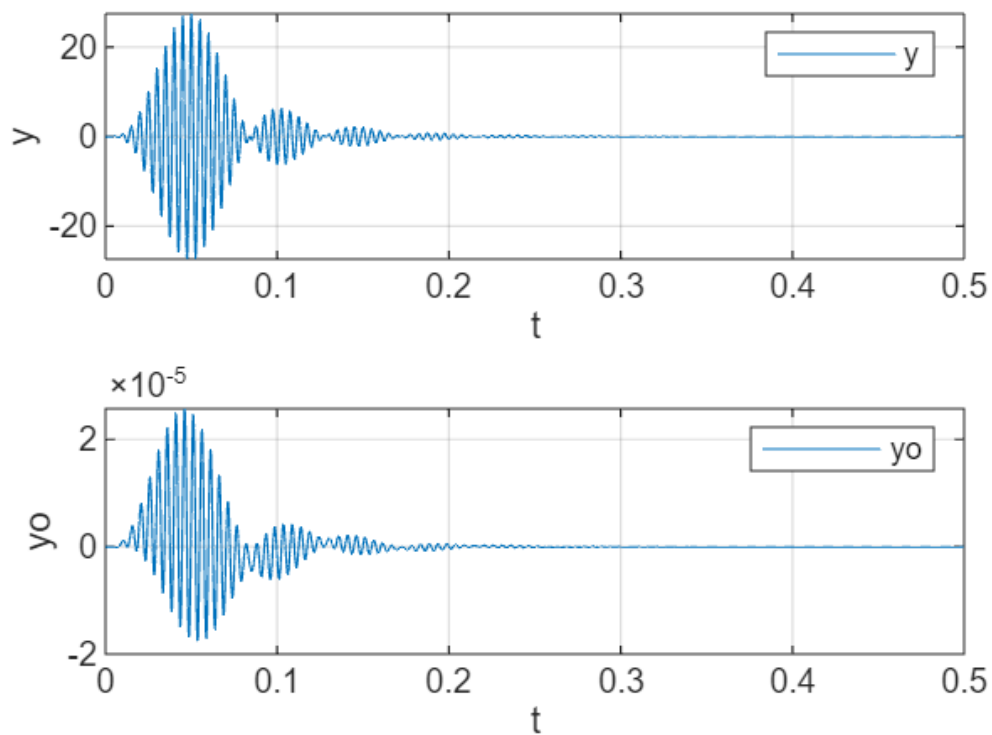
plot(t(1:length(dash)),yo);

xlabel('t'),

ylabel('yo'),

legend('yo'),

grid on;
```



lab result:

The signal waveform is not changed much, but the energy is basically completely lost.

4.6 e

Intermediate Problems

(e). Determine analytically the Fourier transform of each of the signals

$$m(t) \cos(2\pi f_1 t) \cos(2\pi f_1 t),$$

$$m(t) \cos(2\pi f_1 t) \sin(2\pi f_1 t),$$

and

$$m(t) \cos(2\pi f_1 t) \cos(2\pi f_2 t),$$

in terms of $M(j\omega)$, the Fourier transform of $m(t)$.

They ask you to find the Fourier transform of three functions analytically

Analysis:

Just solve

This problem needs no code

1. $m(t) \cos(2\pi f_1 t) \cos(2\pi f_1 t)$:

Using the identity $\cos(a) \cos(b) = \frac{1}{2} [\cos(a+b) + \cos(a-b)]$:

$$\begin{aligned} m(t) \cos^2(2\pi f_1 t) &= m(t) \frac{1}{2} [1 + \cos(4\pi f_1 t)] \\ &= \frac{1}{2} m(t) + \frac{1}{2} m(t) \cos(4\pi f_1 t) \end{aligned}$$

Fourier Transform:

$$\frac{1}{2} M(j\omega) + \frac{1}{2} [M(j(\omega - 4\pi f_1)) + M(j(\omega + 4\pi f_1))]$$

2. $m(t) \cos(2\pi f_1 t) \sin(2\pi f_1 t)$:

Using the identity $\cos(a) \sin(b) = \frac{1}{2} [\sin(a+b) - \sin(a-b)]$:

$$m(t) \cos(2\pi f_1 t) \sin(2\pi f_1 t) = \frac{1}{2} m(t) [\sin(4\pi f_1 t)]$$

Fourier Transform:

$$\frac{1}{2j} [M(j(\omega - 4\pi f_1)) - M(j(\omega + 4\pi f_1))]$$

3. $m(t) \cos(2\pi f_1 t) \cos(2\pi f_2 t)$:

Again using $\cos(a) \cos(b) = \frac{1}{2} [\cos(a+b) + \cos(a-b)]$:

$$m(t) \cos(2\pi f_1 t) \cos(2\pi f_2 t) = \frac{1}{2} m(t) [\cos(2\pi(f_1 + f_2)t) + \cos(2\pi(f_1 - f_2)t)]$$

Fourier Transform:

$$\frac{1}{2} [M(j(\omega - 2\pi(f_1 + f_2))) + M(j(\omega + 2\pi(f_1 + f_2))) + M(j(\omega - 2\pi(f_1 - f_2))) + M(j(\omega + 2\pi(f_1 - f_2)))]$$

lab result:

The Fourier transform is shown in the figure

4.6 f

(f). Using your results from Part (e) and by examining the frequency response of the Alter as plotted in Part (b), devise a plan for extracting the signal $m_1(t)$ from $x(t)$. Plot the signal $m_1(t)$ and determine which letter is represented in Morse code by the signal.

The problem asks the design method to extract $m_1(t)$ and design $m_1(t)$ to identify the letter corresponding to the Morse code of this signal.

Analysis:

From the previous question, we can see that $m_1(t)$ in $x(t)$ is modulated by $\cos(2\pi f_1 t)$. From the above question, we can see that after multiplying it with the same signal again, the low frequency region can be obtained.

The original signal with the current amplitude halved. Therefore, there is a modulated signal modulating a modulated low-pass filter and the original signal can be restored by doubling the amplitude.

```
clear;
```

```
clc;

load newctftmod.mat;

m1t = x.*cos(2*pi*f1*t);

m1 = 2*lsim(bf,af,m1t,t);

figure

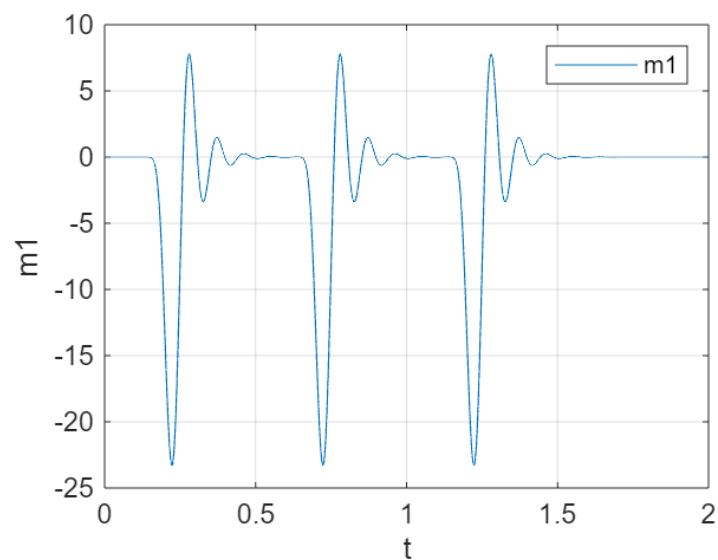
plot(t,m1);

xlabel('t'),

ylabel('m1'),

legend('m1'),

grid on;
```



lab result:

From the textbook diagram, we can see that the Moss code stands for S

4.6 g

(g). Repeat Part (f) for the signals $m_2(t)$ and $m_3(t)$. Agent 008, where does the future of technology lie?

The question asks to reproduce $m_2(t)$ and $m_3(t)$ in the same way, and to restore the last words of Agent 008.

Analysis:

Same as f.

```
clear;

clc;

load newctftmod.mat;

m1t = x.*cos(2*pi*f1*t);

m1 = 2*lsim(bf,af,m1t,t);

figure

plot(t,m1);

xlabel('t'),

ylabel('m1'),

legend('m1'),

grid on;

m2t = x.*sin(2*pi*f2*t);

m2 = 2*lsim(bf,af,m2t,t);

figure
```

```
plot(t,m2);

xlabel('t'),

ylabel('m2'),

legend('m2'),

grid on;

m3t = x.*sin(2*pi*f1*t);

m3 = 2*lsim(bf,af,m3t,t);

figure

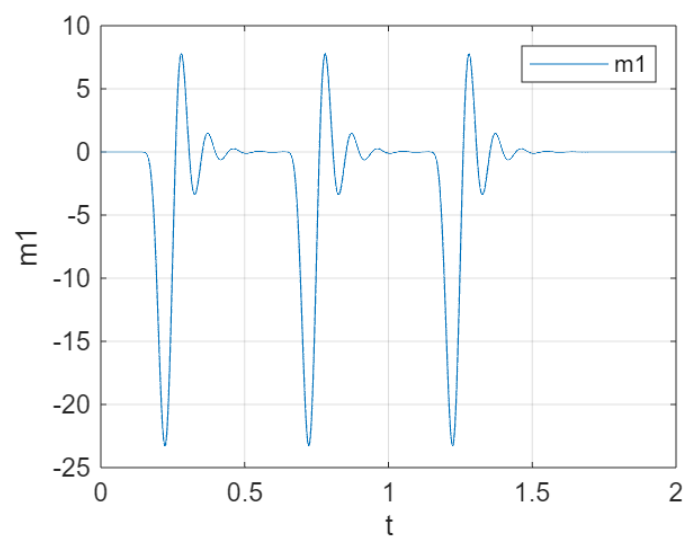
plot(t,m1);

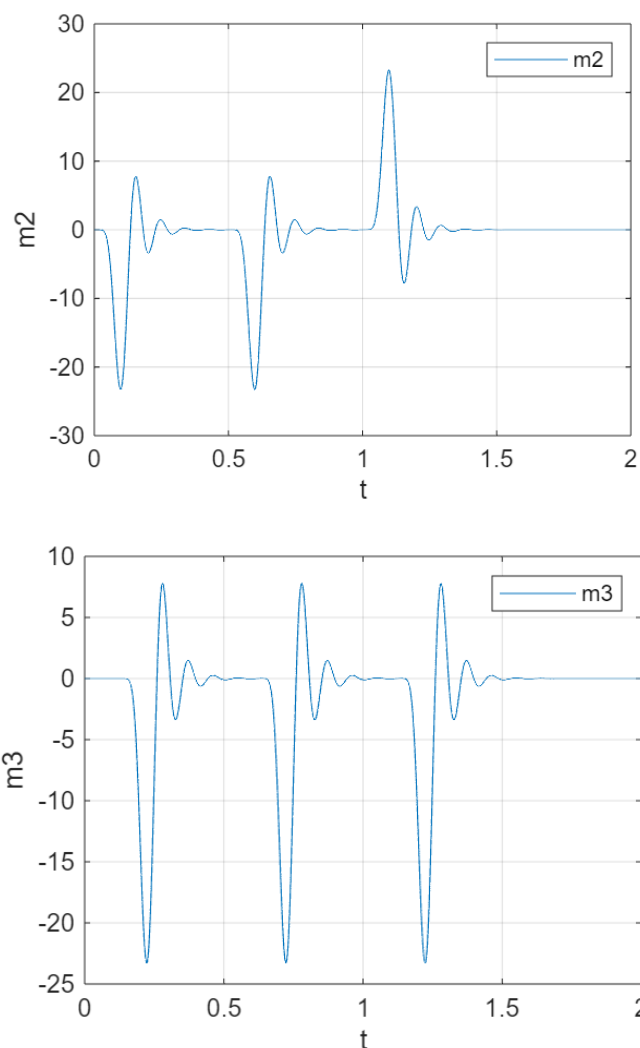
xlabel('t'),

ylabel('m3'),

legend('m3'),

grid on;
```





lab result:

After comparison, the Moss codes corresponding to m_2 and m_3 are ..- And..., correspond to The letters U and S , so Agent008's last words are: "The future of technology lies in **SUS**." The future of technology lies in SUSTech

Note: Please indicate meaning of the symbols in all expressions. Please indicate the coordinate and unit in all figures.

Experience 1. The mathematical foundation required for understanding the CTFT can be quite complex, especially for those new to signal processing. To address this, we will focus on building a strong conceptual understanding through step-by-step tutorials and hands-on exercises. Engaging with additional resources such as instructional videos, and collaborative study sessions can also help in demystifying the more intricate parts of the theory. 2. Applying theoretical concepts of the CTFT to real-world problems can be challenging due to the abstract nature of frequency-domain analysis. To bridge this gap, we will use simulation tools and software to visualize the effects of transformations and to conduct practical experiments. This hands-on approach will help us see the direct impact of theoretical principles in practical scenarios, enhancing our learning experience. 3. The use of software tools and functions like <code>freqs</code> and <code>fourier</code> for computing CTFTs and analyzing system responses requires some familiarity with numerical computation. To overcome this challenge, we will provide detailed guidelines and practice exercises focusing on these tools. Learning these computational techniques in a step-by-step manner will equip us with the necessary skills to effectively use technology in analyzing and solving signal processing problems.	
Score	100/100

字体：英文 Times new Roman;